# Predicting Vortex Shedding Behind a Circular Cylinder with Physics-Informed Neural Networks

Peeyush Agarwal
Department of Aerospace Engineering
Indian Institute of Technology Kanpur
Kanpur, India

***Abstract*—Physics-informed neural networks (PINNs) trained without data on the unsteady wake of a circular cylinder tend toward the steady symmetric solution, which satisfies the Navier–Stokes equations exactly. We study this failure at $\mathrm{Re} = 100$ using lattice Boltzmann reference data validated against Qu et al. ($\mathrm{St}$ within $0.04\%$, $C_d$ $3.5\%$ high). Averaged over four start phases and five seeds, a data-free PINN retains 31.1% of the reference wake amplitude. Causal time-marching raises this to 56.4%, but at equal training budget it adds only 7.7 points. Combining marching with random Fourier features and a Reynolds decomposition gives 72.9%, between 70% and 77% at every start phase, at a relative $L_2$ error of 0.031. This configuration predicts one shedding period from physics alone, but over five periods it sheds 10.4% slow and ends about half a cycle out of phase. Holding the phase requires 350 supervised points per period taken from the predicted span, which turns the task into reconstruction from sparse measurements. The gain from the combined strategies holds for $\mathrm{Re} = 60$–$180$. A recurrent sequence PINN does not replace marching. For data-driven forecasting, a physics residual makes a Fourier neural operator worse on the validated data; on an earlier, harder dataset it helps the operator at short horizons, but not a U-Net or a convolutional GRU.**

***Index Terms*—physics-informed neural networks, vortex shedding, causal training, lattice Boltzmann method, Fourier neural operator**

## I. Introduction

Above $\mathrm{Re} \approx 47$ the wake of a circular cylinder sheds a Kármán vortex street at a frequency given by the Strouhal number $\mathrm{St} = fD/U$. Predicting this flow with a data-free PINN is difficult for three reasons. The steady symmetric field satisfies the equations and boundary conditions exactly, so it is an easily reached minimum of the residual loss. Networks learn low frequencies first (spectral bias), which works against an oscillatory wake. The initial condition is a single time slice and is weakly weighted against the residual over the space–time domain.

Chuang and Barba [1] found that a data-free PINN behaves like a steady solver, and Shah [2], on a confined cylinder benchmark, found that only supervised combinations of training techniques escape this collapse. Causal training, either by time-marching [3] or by causal weighting [4], has been shown to help on one-dimensional problems, but neither of these cylinder studies tests hard time-window marching. Separately, Renn et al. [5] forecast cylinder wakes with a purely data-driven Fourier neural operator (FNO) [6].

TABLE I
Validation at $\mathrm{Re} = 100$ against Qu et al. [8]

| Quantity | Present | Benchmark range | Error |
|---|---|---|---|
| Mean drag $C_d$ | 1.3634 | 1.318–1.336 | $+3.5\%$ |
| Strouhal number St | 0.1648 | 0.164–0.166 | $-0.04\%$ |
| R.m.s. lift $C_l'$ | 0.2362 | 0.222–0.235 | $+6.2\%$ |

Errors are relative to their values $C_d = 1.317$, $\mathrm{St} = 0.1649$ and $C_l' = 0.2224$; the range is their compilation of eight studies.

The objective of this work is to reproduce the PINN collapse on validated reference data, to identify which training strategy prevents it under a controlled comparison, and to test whether adding physics to an operator forecast improves it. Here, prediction means computing the flow beyond the data the model is given; the reference data are otherwise used only for scoring. The contributions are an unsteady Navier–Stokes operator for the `underPINN` framework [7], a validated lattice Boltzmann dataset, a phase-averaged ablation with equal-budget controls, and a comparison of physics-informed forecasting across three operator backbones.

## II. Reference Data

The reference wake is computed with a D2Q9 two-relaxation-time lattice Boltzmann solver written in JAX, with half-way bounce-back on the cylinder and free-slip lateral walls. The inlet is $25D$ upstream of the cylinder in a domain $60D$ long. The production run uses $\mathrm{Re} = 100$, $D = 24$ lattice units, $\mathrm{Ma} = 0.087$, a $1440 \times 1200$ grid and $2\%$ blockage. It records 641 snapshots over 20 shedding periods on a $241 \times 121$ grid ($\Delta x = 0.083D$, $\Delta t^* = 0.19$). Table I compares it with Qu et al. [8].

### A. Validation of the Reference Solution

Two corrections were needed. First, the free-slip walls reflect sound, so the domain behaves as an acoustic duct with a transverse mode at $\mathrm{St}_a = 0.1155$. Without treatment the wake locks onto this mode over long records ($\mathrm{St} = 0.1132$ over 38 periods). An absorbing layer of $8D$ at each wall that relaxes the populations toward the free-stream equilibrium removes it. Blevins [9] showed experimentally that a transverse acoustic wave can shift cylinder shedding to its own frequency. Second, at $D = 32$, moving the inlet from $8D$ to $25D$ upstream lowers $C_d$ by $4.7\%$, from $+7.4\%$ to $+2.3\%$ against Qu et al., and

takes St from $+3.0\%$ to $+0.3\%$; the domain height has a much smaller effect. The grid is converged at $D = 32$. The production value $D = 24$ reads $C_d$ about $1.4\%$ high and is kept so that the snapshot spacing matches PIV-like resolution. None of the PINN or operator results depend on $C_d$.

## III. Methodology

### A. Unsteady Navier–Stokes Operator

The network maps $(x, y, t) \mapsto (u, v, p)$ and is trained on the non-dimensional incompressible equations

$$u_x + v_y = 0, \tag{1}$$

$$u_t + uu_x + vu_y + p_x - \tfrac{1}{\mathrm{Re}}(u_{xx} + u_{yy}) = 0, \tag{2}$$

$$v_t + uv_x + vv_y + p_y - \tfrac{1}{\mathrm{Re}}(v_{xx} + v_{yy}) = 0. \tag{3}$$

`underPINN` provides only a steady residual, so an unsteady operator was implemented, with derivatives computed by forward-mode automatic differentiation. It annihilates the Taylor–Green vortex to $1.6 \times 10^{-16}$, matches a reference Hessian implementation to $2.8 \times 10^{-17}$, and reduces to the framework's steady residual to $5.6 \times 10^{-17}$.

### B. Training Strategies

The baseline is a multilayer perceptron $[3, 128^{\times 6}, 3]$ trained for 20 000 epochs on the equation residual, boundary conditions and initial condition over one shedding period, with no data other than the initial condition. Three strategies are added as independent switches on the same trainer:

- *Fourier features* [10], against spectral bias. The bandwidth is $\sigma = 8$, except in the phase-0 ablation and the supervision study, which use $\sigma = 2$.
- *Reynolds decomposition* $u = u_0 + u'$, where the mean $u_0$ is a separate frozen network fitted to the measured mean.
- *Causal time-marching*: four windows of a quarter period solved in sequence, each taking its initial condition from the previous window's network. The run uses ground truth only once.

The combination of all three is denoted `all`.

### C. Evaluation Protocol

The main metric is *shedding retained*: the ratio of transverse-velocity r.m.s. at a wake probe between the PINN and the reference. A value near 0% means collapse to steady flow. Field accuracy is the relative $L_2$ error of the velocity field. Because retention depends on the phase of the cycle at which a run starts, the main comparison is averaged over four start phases with five seeds each. A marched run trains four windows, so each marched configuration is also compared with a single-window run given the same 80 000 epochs.

## IV. Results

### A. Baseline PINN

The baseline retains between 19.0% and 43.2% of the wake depending on the start phase, 31.1% on average (Table II). Training for 80 000 epochs raises the mean to 48.6% but leaves the dependence on start phase in place.

TABLE II
Shedding Retained by Start Phase (Five Seeds Each)

| Configuration | 0 | $\frac{1}{4}$ | $\frac{1}{2}$ | $\frac{3}{4}$ | Mean | $L_2$ |
|---|---|---|---|---|---|---|
| Baseline | 43.2% | 20.3% | 42.1% | 19.0% | 31.1% | 0.166 |
| Marching | 70.6% | 40.4% | 73.8% | 40.7% | 56.4% | 0.156 |
| `all` | 74.9% | 70.2% | 76.6% | 70.0% | **72.9%** | **0.031** |

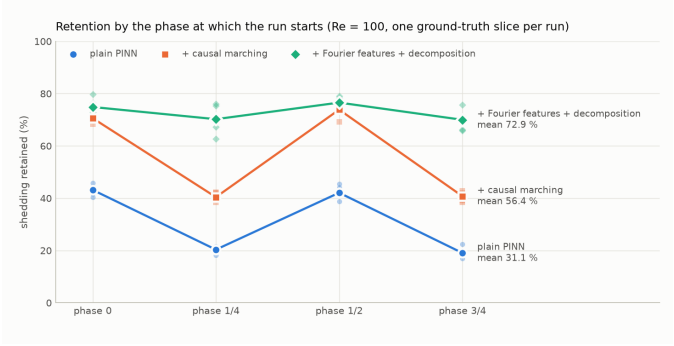



Fig. 1. Shedding retained by start phase, all runs.

TABLE III
Marching Against Single-Window Runs of Equal Budget

| Comparison | Marched | Single | Gain | Paired range |
|---|---|---|---|---|
| Marching vs baseline | 56.4% | 48.6% | +7.7 | $-2.2$ to $+15.8$ |
| `all` vs F+D | 72.9% | 38.7% | +34.1 | $+20.0$ to $+55.2$ |

80 000 epochs for both; averaged over four start phases and five seeds; gains in points, paired by phase and seed. F+D: Fourier features and Reynolds decomposition.

### B. Ablation of Training Strategies

In a phase-0 ablation (two seeds), Fourier features (36.9%) and the Reynolds decomposition (36.1%) each retain slightly less than the baseline (41.1%), although the decomposition halves the field error. Marching raises retention at every start phase, from 31.1% to 56.4% on average, but keeps the baseline's dependence on start phase. Only the combination of all three is insensitive to start phase, at a fifth of the field error of either alternative (Table II, Fig. 1). The 95% intervals over seeds are $\pm 1.5$, $\pm 0.9$ and $\pm 4.6$ points for the three rows.

Table III separates the effect of marching from the extra training it receives. Alone, marching beats an equal-budget single window by 7.7 points, less than the 17.5 points that longer training gives the baseline. With Fourier features and the decomposition in place, marching adds 34.1 points and wins all 20 pairings. The strategies are therefore complementary, in agreement with Shah [2]. An earlier version of this study gave each marched window a fresh reference snapshot; that raised retention at phase 0 to 93.0% and 94.5%, so about twenty points of the earlier result came from extra ground truth.

### C. Prediction Horizon and Sparse Measurements

Over one period, `all` matches the reference phase to $0.1°$, at 73.5% of its amplitude. Over five chained periods it keeps

TABLE IV
SPARSE RE-ANCHORING OF THE FIVE-PERIOD CHAIN

| $N$ | $K$ | Per period | Total | St error (%) | Lag |
|---|---|---|---|---|---|
| — | — | 0 | 0 | −10.4, −10.2 | 0.369, 0.345 |
| 100 | 2 | 200 | 1 000 | −4.6, −3.3 | 0.120, 0.100 |
| 150 | 2 | 300 | 1 500 | −1.5, −1.3 | 0.053, 0.042 |
| 175 | 2 | 350 | 1 750 | −0.2, −0.7 | 0.002, 0.002 |
| 100 | 1 | 400 | 2 000 | −0.3, −0.2 | 0.001, 0.001 |

Selected rows, two seeds each. Lag is the cross-correlation lag over the last period, in shedding periods, positive behind the reference.

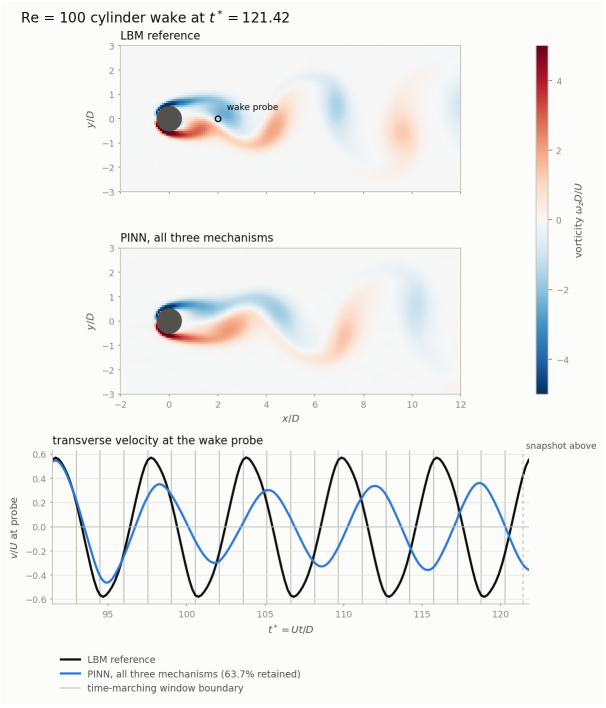



Fig. 2. Data-free `all` marched over five periods against the reference. Top: vorticity at the end of the run. Bottom: transverse velocity at the wake probe; grey lines are window boundaries.

shedding, but at $\mathrm{St} = 0.1476$, 10.4% below the reference, and ends about half a cycle out of phase (Fig. 2); amplitude falls to 63.8% and the field error rises to 0.156. Data-free marching therefore predicts about one period and sustains a wake beyond it, but does not hold the phase.

Measurements from inside the predicted span correct this. With supervised points in every window (start phase 0, $\sigma = 2$, two seeds), 100 points per window give 87.4% retention and 500 give 96.9%, with the field error falling to 0.010. Table IV gives selected rows of a sparse re-anchoring study, in which $N$ supervised points are given to every $K$-th window of the five-period chain and initial conditions stay chained. The phase locks between 300 and 350 points per period: at 350 per period, 1 750 points in total and about 2% of what per-window snapshots would use, both seeds hold the phase to within 0.003 of a period and St to within 0.7%. Because these points lie inside the span being computed, this result is reconstruction from sparse measurements, not forecasting.

TABLE V
DATA-FREE RESULTS ACROSS REYNOLDS NUMBER

| | | Shedding retained | | $L_2$ | |
|---|---|---|---|---|---|
| Re | St error | Baseline | `all` | Base. | `all` |
| 60 | +0.51% | 29.8, 33.5% | 62.8, 69.1% | 0.163 | 0.028 |
| 80 | +0.60% | 40.2, 44.6% | 75.5, 76.9% | 0.173 | 0.028 |
| 100 | −0.04% | 38.8, 41.5% | 71.8, 78.5% | 0.173 | 0.031 |
| 120 | +0.03% | 23.9, 24.1% | 71.5, 78.6% | 0.170 | 0.036 |
| 150 | −0.28% | 38.0, 41.4% | 69.7, 75.1% | 0.174 | 0.039 |
| 180 | −0.45% | 24.1, 27.7% | 77.5, 76.6% | 0.171 | 0.040 |

St error is against Qu et al.; retention for each of two seeds; $L_2$ is the two-seed mean.

### D. Reynolds-Number Dependence

The comparison was repeated at $\mathrm{Re} = 60$–180, with new reference data at each Reynolds number; the upper limit is set by the three-dimensional transition at $\mathrm{Re}_c = 188.5$ [11]. Every dataset's St is within 0.6% of Qu et al. The baseline retains 24–45% and `all` 63–79%, and `all` reduces the field error from about 0.17 to 0.028–0.040 (Table V). These runs use two seeds and a single start phase, the beginning of each record, so the variation in the baseline column reflects start phase and has no physical meaning. The insensitivity of `all` to start phase was measured only at $\mathrm{Re} = 100$.

### E. Sequence Architectures

PINNsFormer [12] and a GRU variant with the same embedding were compared with the MLP. The GRU and MLP were run with two seeds at all four start phases. On a single window the GRU retains 21.8% against the MLP's 30.1%, and marched it retains 33.6% against 56.7%. PINNsFormer was run once, at phase 0, because of its cost: on a single window it retains 34.9% against the MLP's 40.4%, and marched it reaches 77.5% against 69.7%, at 11.8 times the MLP's run time. Neither architecture prevents the collapse without marching, and one marched run is not enough to show that attention adds to it.

### F. Physics-Informed Operator Forecasting

Following Renn et al. [5], an FNO maps two consecutive snapshots to the next and is rolled out recursively. It trains on the first 70% of the record and forecasts from states in the last 30%, so this is true forecasting. A physics term is added through the vorticity-transport and continuity residuals, which do not require pressure. On the validated dataset the data-driven FNO reaches an error of 0.0010 at its trained horizon of ten steps and 0.0152 after 50 steps, and the physics term makes it 16–18% worse at ten and twenty steps. Test states on this clean limit cycle are close to the training data, which may explain why physics does not help.

On an earlier dataset whose fields contain the acoustic mode, and whose held-out states are therefore less like the training data, physics lowers the FNO error by 21% at ten steps and 28% at twenty, then crosses over at 0.8–1.0 shedding periods. Table VI repeats this test for a U-Net [13] and a convolutional GRU [14]. The physics term changes the U-Net

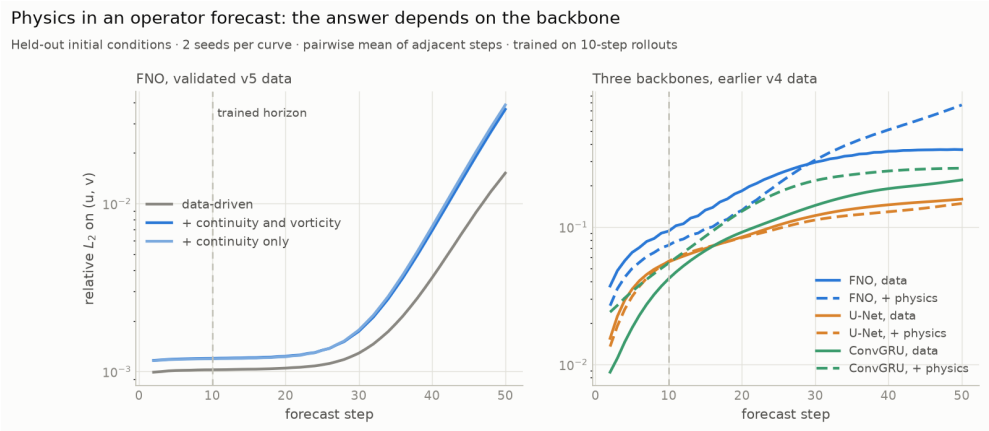



Fig. 3. Forecast error against horizon. Left: FNO on the validated dataset. Right: three backbones on the earlier dataset, data-driven (solid) and with physics (dashed).

TABLE VI
FORECAST ERROR ON THE EARLIER DATASET

| | | Data-driven error | | | Change with physics | | |
|---|---|---|---|---|---|---|---|
| Backbone | Params | 10 | 20 | 50 | 10 | 20 | 50 |
| FNO | 59.0 M | 0.093 | 0.183 | 0.365 | −21% | −28% | +111% |
| U-Net | 1.93 M | 0.056 | 0.084 | 0.159 | +1% | −1% | −7% |
| ConvGRU | 0.45 M | 0.042 | 0.092 | 0.220 | +30% | +42% | +22% |

Columns are forecast steps; negative change is better. Two seeds.

error by at most 7% and makes the ConvGRU worse, so the short-horizon benefit is specific to the FNO (Fig. 3). Without physics, both convolutional networks forecast better than the FNO, with 31 and 131 times fewer parameters.

## V. CONCLUSION

A data-free PINN retains 31.1% of the cylinder wake at $\mathrm{Re} = 100$, and the amount depends on the start phase. No single strategy fixes this. Causal time-marching adds only 7.7 points over an equal-budget baseline, but combined with Fourier features and a Reynolds decomposition it retains 72.9% on average and between 70% and 77% at every start phase, at a field error of 0.031, with a similar gain across $\mathrm{Re} = 60$–$180$. From physics alone, this PINN predicts about one shedding period; over five it drifts in phase. About 350 supervised points per period from inside the predicted span hold the phase, which is reconstruction rather than forecasting. Adding physics to an operator forecast helped only the FNO, only at short horizons, and only on the earlier dataset.

The main limitations are that most secondary studies use two seeds and PINNsFormer was run once, that continuity between marching windows is enforced only through the loss (exact enforcement [15] is a possible next step), and that the backbone comparison was run only on the earlier dataset.

## REFERENCES


[1] P.-Y. Chuang and L. A. Barba, "Predictive limitations of physics-informed neural networks in vortex shedding," arXiv:2306.00230, 2023.

[2] D. Shah, "Complementary, not cumulative: Interaction effects in physics-informed neural networks for Navier–Stokes vortex shedding," arXiv:2608.19632, 2026.

[3] A. S. Krishnapriyan, A. Gholami, S. Zhe, R. M. Kirby, and M. W. Mahoney, "Characterizing possible failure modes in physics-informed neural networks," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2021.

[4] S. Wang, S. Sankaran, and P. Perdikaris, "Respecting causality is all you need for training physics-informed neural networks," *Comput. Methods Appl. Mech. Eng.*, 2024, doi: 10.1016/j.cma.2024.116813.

[5] P. J. Renn, C. Wang, S. Lale, Z. Li, A. Anandkumar, and M. Gharib, "Forecasting subcritical cylinder wakes with Fourier neural operators," arXiv:2301.08290, 2023.

[6] Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, and A. Anandkumar, "Fourier neural operator for parametric partial differential equations," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021.

[7] Aeroscience Computations and Analysis Lab, "underPINN: A JAX framework for physics-informed neural networks." [Online]. Available: https://github.com/Aeroscience-Computations-Analysis-Lab/underPINN

[8] L. Qu, C. Norberg, L. Davidson, S.-H. Peng, and F. Wang, "Quantitative numerical analysis of flow past a circular cylinder at Reynolds number between 50 and 200," *J. Fluids Struct.*, vol. 39, pp. 347–370, 2013.

[9] R. D. Blevins, "The effect of sound on vortex shedding from cylinders," *J. Fluid Mech.*, vol. 161, pp. 217–237, 1985.

[10] M. Tancik *et al.*, "Fourier features let networks learn high frequency functions in low dimensional domains," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2020.

[11] D. Barkley and R. D. Henderson, "Three-dimensional Floquet stability analysis of the wake of a circular cylinder," *J. Fluid Mech.*, vol. 322, pp. 215–241, 1996.

[12] Z. Zhao, X. Ding, and B. A. Prakash, "PINNsFormer: A transformer-based framework for physics-informed neural networks," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2024.

[13] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Med. Image Comput. Comput. Assist. Interv. (MICCAI)*, LNCS vol. 9351, 2015, pp. 234–241.

[14] P. Ren, C. Rao, Y. Liu, J.-X. Wang, and H. Sun, "PhyCRNet: Physics-informed convolutional-recurrent network for solving spatiotemporal PDEs," *Comput. Methods Appl. Mech. Eng.*, vol. 389, Art. no. 114399, 2022.

[15] P. Roy and S. Castonguay, "Exact enforcement of temporal continuity in sequential physics-informed neural networks," arXiv:2403.03223, 2024.